Database Systems
CSE 215

Lecture 03: SQL

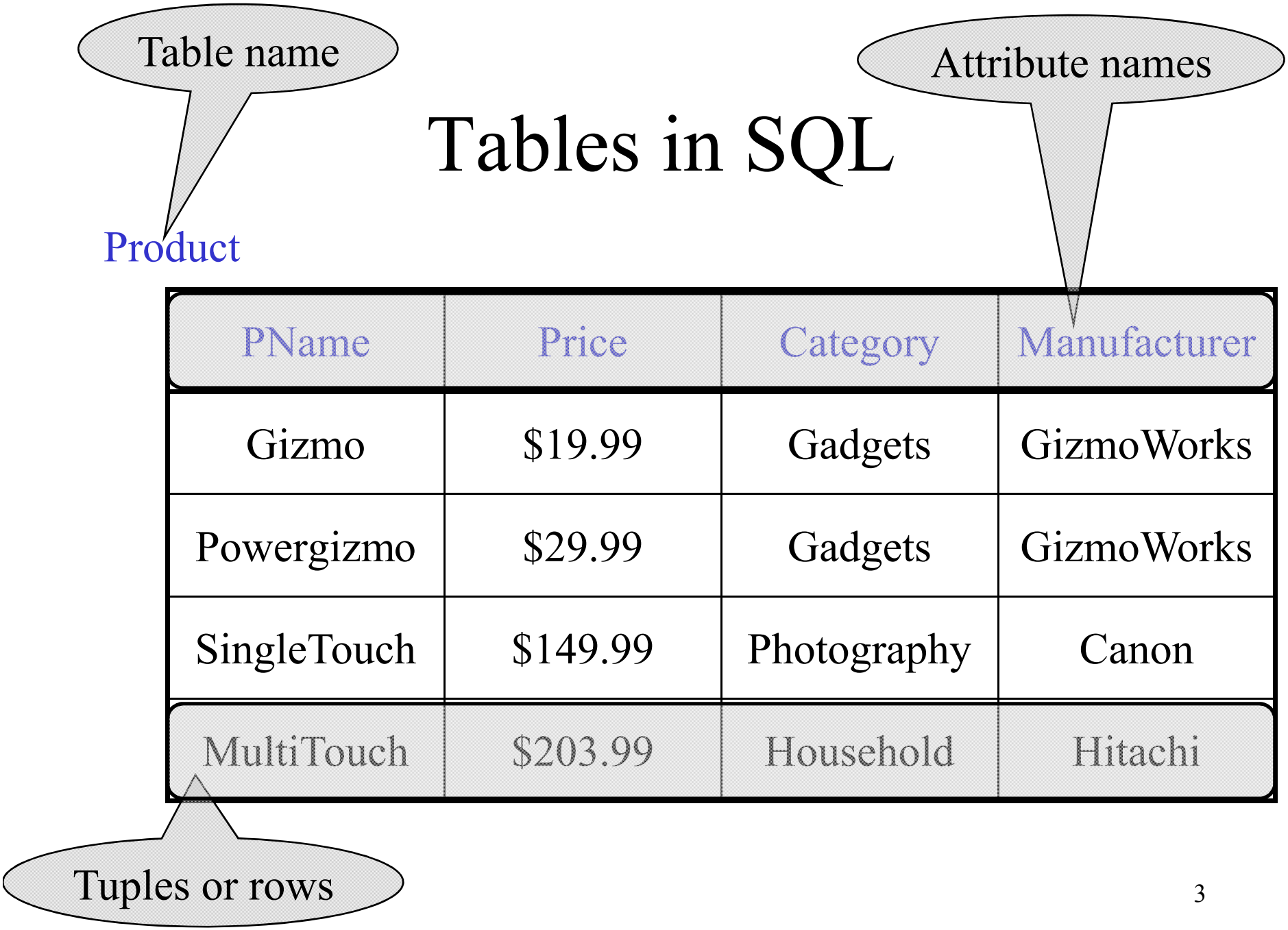
2023

Single Table Query

SQL

- Data Definition Language (DDL)
  - Create/alter/delete tables and their attributes
  - Previous lectures...
- Data Manipulation Language (DML)
  - Query one or more tables – discussed next !
  - Insert/delete/modify tuples in tables

Tables in SQL



SQL Query

Basic form: (plus many many more bells and whistles)

```
SELECT  <attributes>
FROM    <one or more relations>
```

Simplest SQL Query

Product

PName	Price	Category	Manufacturer
Gizmo	$19.99	Gadgets	GizmoWorks
Powergizmo	$29.99	Gadgets	GizmoWorks
SingleTouch	$149.99	Photography	Canon
MultiTouch	$203.99	Household	Hitachi

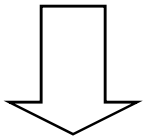
Show all entries from product table

Simplest SQL Query

Product

PName	Price	Category	Manufacturer
Gizmo	$19.99	Gadgets	GizmoWorks
Powergizmo	$29.99	Gadgets	GizmoWorks
SingleTouch	$149.99	Photography	Canon
MultiTouch	$203.99	Household	Hitachi

```
SELECT  *
FROM    Product
```



PName	Price	Category	Manufacturer
Gizmo	$19.99	Gadgets	GizmoWorks
Powergizmo	$29.99	Gadgets	GizmoWorks
SingleTouch	$149.99	Photography	Canon
MultiTouch	$203.99	Household	Hitachi

"selection"

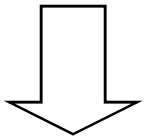
Single Column Selection

Show all product categories

Product

PName	Price	Category	Manufacturer
Gizmo	$19.99	Gadgets	GizmoWorks
Powergizmo	$29.99	Gadgets	GizmoWorks
SingleTouch	$149.99	Photography	Canon
MultiTouch	$203.99	Household	Hitachi

```
SELECT   category
FROM     Product
```



Category
Gadgets
Gadgets
Photography
Household

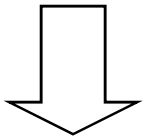
"selection" and "projection"

Single Column Selection (Eliminating Duplicates)

Product

PName	Price	Category	Manufacturer
Gizmo	$19.99	Gadgets	GizmoWorks
Powergizmo	$29.99	Gadgets	GizmoWorks
SingleTouch	$149.99	Photography	Canon
MultiTouch	$203.99	Household	Hitachi

```
SELECT  DISTINCT  category
FROM    Product
```

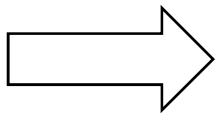


Category
Gadgets
Photography
Household

Show all product categories

In summary..

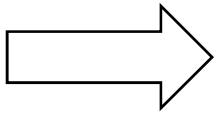
```
SELECT   DISTINCT category
FROM     Product
```



Category
Gadgets
Photography
Household

Compare to:

```
SELECT   category
FROM     Product
```



Category
Gadgets
Gadgets
Photography
Household

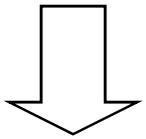
Multi-column Selection

List all products along with their price and manufacturers

Product

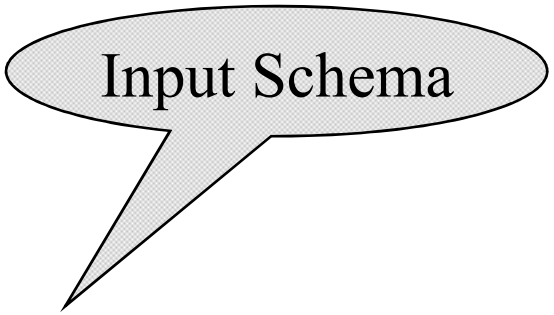
PName	Price	Category	Manufacturer
Gizmo	$19.99	Gadgets	GizmoWorks
Powergizmo	$29.99	Gadgets	GizmoWorks
SingleTouch	$149.99	Photography	Canon
MultiTouch	$203.99	Household	Hitachi

```
SELECT  pname, price, manufacturer
FROM    Product
```



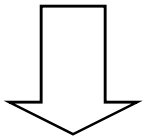
PName	Price	Manufacturer
Gizmo	$19.99	GizmoWorks
Powergizmo	$29.99	GizmoWorks
SingleTouch	$149.99	Canon
MultiTouch	$203.99	Hitachi

Notation

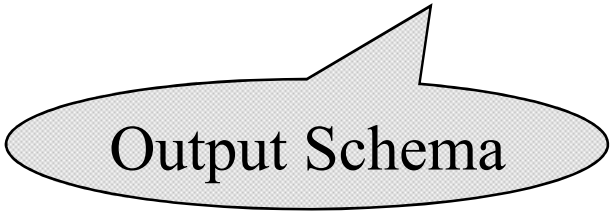


Product(PName, Price, Category, Manfacturer)

```
SELECT   pname, price, manufacturer
FROM     Product
```



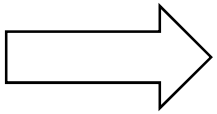
Answer(PName, Price, Manfacturer)



Duplicates from multiple column

Maker	Model	Type
A	1001	pc
A	1002	pc
A	1003	pc
A	2004	laptop
A	2005	laptop
A	2006	laptop
B	1004	pc
B	1005	pc
B	1006	pc
B	2007	laptop
C	1007	pc
…	…	…
….	…	…

```
SELECT  maker, type
FROM    Product
```



?

Duplicates from multiple column

Maker	Type
A	pc
A	pc
A	pc
A	laptop
A	laptop
A	laptop
B	pc
B	pc
B	pc
B	laptop
C	pc
…	…
….	…

```
SELECT   maker, type
FROM     Product
```

Duplicates from multiple column

Maker	Type
A	pc
A	pc
A	pc
A	laptop
A	laptop
A	laptop
B	pc
B	pc
B	pc
B	laptop
C	pc
…	…
….	…

```
SELECT DISTINCT   maker, type
FROM      Product
```

Duplicates from multiple column

Maker	Type
A	pc
A	laptop
B	pc
B	laptop
C	pc
…	…
….	…
…	…
….	…
…	…
….	…
…	…
….	…

Simple Predicates

3 SELECT <attributes>
1 FROM <one or more relations>
2 WHERE <conditions>

Semantics

3 SELECT $a_1,\ldots,a_k$
1 FROM R
2 WHERE conditions

```
Answer = {}
for each x in R do
        if conditions satisfied then
                Answer = Answer + {(a1,…,ak)}
return Answer
```

Semantics

3 SELECT DISTINCT $a_1,\ldots,a_k$
1 FROM R
2 WHERE conditions

```
Answer = {}
for each x in R do
        if conditions satisfied then
                Answer = Answer ∪ {(a1,…,ak)}
return Answer
```

Number Range Selection

Show all product's info that will cost more than 100 dollars

Product

PName	Price	Category	Manufacturer
Gizmo	$19.99	Gadgets	GizmoWorks
Powergizmo	$29.99	Gadgets	GizmoWorks
SingleTouch	$149.99	Photography	Canon
MultiTouch	$203.99	Household	Hitachi

```
SELECT  *
FROM    Product
WHERE   price > 100
```

```
Answer = {}
for each x in R do
        if conditions satisfied then
                Answer = Answer + {(a1,…,ak)}
return Answer
```

PName	Price	Category	Manufacturer
SingleTouch	$149.99	Photography	Canon
MultiTouch	$203.99	Household	Hitachi

Number Range Operators

- Greater than “>”
- Less than “<”
- Greater than or equal to “>=”
- Less than or equal to “<=”
- Two forms of not-equal-to

  “!=” and “< >”

BETWEEN-AND operator

```
SELECT <attributes>
FROM   <table>
WHERE  attribute BETWEEN value1 AND value 2
```

BETWEEN-AND
is inclusive

Works on numbers, texts and dates

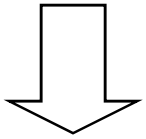
BETWEEN-AND operator

Show all product's info in the price range $100 to $200 (inclusive)

Product

PName	Price	Category	Manufacturer
Gizmo	$19.99	Gadgets	GizmoWorks
Powergizmo	$29.99	Gadgets	GizmoWorks
SingleTouch	$149.99	Photography	Canon
MultiTouch	$203.99	Household	Hitachi

```
SELECT  *
FROM    Product
WHERE   price BETWEEN 100 AND 200
```



PName	Price	Category	Manufacturer
SingleTouch	$149.99	Photography	Canon

BETWEEN-AND is inclusive

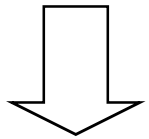
NOT BETWEEN-AND operator

Show all product's info not in the price range $10 to $200 (inclusive)

Product

PName	Price	Category	Manufacturer
Gizmo	$19.99	Gadgets	GizmoWorks
Powergizmo	$29.99	Gadgets	GizmoWorks
SingleTouch	$149.99	Photography	Canon
MultiTouch	$203.99	Household	Hitachi

```
SELECT  *
FROM    Product
WHERE   price NOT BETWEEN 10 AND 200
```



NOT BETWEEN-AND is also inclusive

PName	Price	Category	Manufacturer
MultiTouch	$203.99	Household	Hitachi

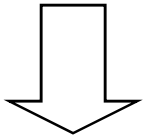
String Matching

Show all household product's info

Product

PName	Price	Category	Manufacturer
Gizmo	$19.99	Gadgets	GizmoWorks
Powergizmo	$29.99	Gadgets	GizmoWorks
SingleTouch	$149.99	Photography	Canon
MultiTouch	$203.99	Household	Hitachi

```
SELECT   pname, price, manufacturer
FROM     Product
WHERE    category ='Household'
```



Matches even category is fixed length CHAR data type right padded with blank

PName	Price	Manufacturer
MultiTouch	$203.99	Hitachi

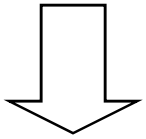
String Matching

Show all household product's info

Product

PName	Price	Category	Manufacturer
Gizmo	$19.99	Gadgets	GizmoWorks
Powergizmo	$29.99	Gadgets	GizmoWorks
SingleTouch	$149.99	Photography	Canon
MultiTouch	$203.99	Household	Hitachi

```
SELECT   pname, price, manufacturer
FROM     Product
WHERE    category ='Household '
```



Matches even category is fixed length CHAR data type right padded with blank but does not match if it's VARCHAR2

PName	Price	Manufacturer
MultiTouch	$203.99	Hitachi

The **LIKE** operator

- s **LIKE** p: pattern matching on strings
- p may contain two special symbols:

  (1) **%** → any sequence of characters (zero or more)

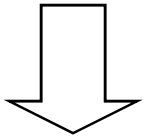
  (2) _ → any single character (exactly one)

The **LIKE** operator

Show all product's info that has 'mo' as a substring of the product name

PName	Price	Category	Manufacturer
Gizmo	$19.99	Gadgets	GizmoWorks
Powergizmo	$29.99	Gadgets	GizmoWorks
SingleTouch	$149.99	Photography	Canon
MultiTouch	$203.99	Household	Hitachi

```
SELECT  *
FROM    Products
WHERE   pname LIKE '%mo%'
```



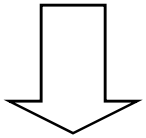
PName	Price	Category	Manufacturer
Gizmo	$19.99	Gadgets	GizmoWorks
Powergizmo	$29.99	Gadgets	GizmoWorks

The **LIKE** operator

Show all product's info where the product name ends with 'Touch'

PName	Price	Category	Manufacturer
Gizmo	$19.99	Gadgets	GizmoWorks
Powergizmo	$29.99	Gadgets	GizmoWorks
SingleTouch	$149.99	Photography	Canon
MultiTouch	$203.99	Household	Hitachi

```
SELECT  *
FROM    Products
WHERE   pname LIKE '%Touch'
```

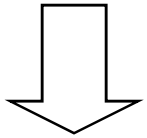


PName	Price	Category	Manufacturer
SingleTouch	$149.99	Photography	Canon
MultiTouch	$203.99	Household	Hitachi

The **LIKE** operator

PName	Price	Category	Manufacturer
Gizmo	$19.99	Gadgets	GizmoWorks
Powergizmo	$29.99	Gadgets	GizmoWorks
SingleTouch	$149.99	Photography	Canon
MultiTouch	$203.99	Household	Hitachi

```
SELECT  *
FROM    Products
WHERE   manufacturer LIKE 'G_z_oWorks'
```



PName	Price	Category	Manufacturer
Gizmo	$19.99	Gadgets	GizmoWorks
Powergizmo	$29.99	Gadgets	GizmoWorks

Searching for three special symbols

- ’ → ‘Baby’s World’
- % → ‘10%’
- _ → ‘USA_Canada’

Searching for three special symbols

- ’ → ‘Baby’s World’

Search for all company names having apostrophe s in the name?

Use double apostrophe

```
SELECT *
FROM Company
WHERE cname LIKE ‘%’’s%’
```

Searching for three special symbols

- % → '10%'

Search for all interest rates having percent at the end?

Use ESCAPE characters

```
SELECT *
FROM Bank
WHERE  interestRates LIKE '%\%' ESCAPE '\'
```

Searching for three special symbols

- ‘_’ → ‘USA_Canada’

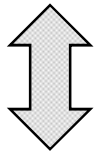
Search for all country names having underscore inside?

Use ESCAPE characters

```
SELECT *
FROM Company
WHERE country LIKE ‘%\_%’ ESCAPE ‘\’
```

LIKE does not pad with blank before matching

```
SELECT   pname, price, manufacturer
FROM     Product
WHERE    category ='Household'
```



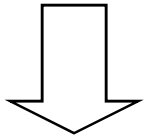
Not equivalent

```
SELECT   pname, price, manufacturer
FROM     Product
WHERE    category LIKE 'Household'
```

The **NOT LIKE** operator

PName	Price	Category	Manufacturer
Gizmo	$19.99	Gadgets	GizmoWorks
Powergizmo	$29.99	Gadgets	GizmoWorks
SingleTouch	$149.99	Photography	Canon
MultiTouch	$203.99	Household	Hitachi

```
SELECT  *
FROM    Products
WHERE   manufacturer NOT LIKE '%mo%'
```



PName	Price	Category	Manufacturer
SingleTouch	$149.99	Photography	Canon
MultiTouch	$203.99	Household	Hitachi

Combining Predicates

Product

PName	Price	Category	Manufacturer
Gizmo	$19.99	Gadgets	GizmoWorks
Powergizmo	$29.99	Gadgets	GizmoWorks
SingleTouch	$149.99	Photography	Canon
MultiTouch	$203.99	Household	Hitachi

Show all household products having prices more than $100

```
SELECT  *
FROM    Products
WHERE   category = 'Household'
AND     price >= 100
```

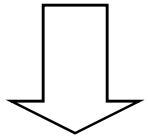
Combining Predicates

Show all manufacturer that make photography or household items

Product

PName	Price	Category	Manufacturer
Gizmo	$19.99	Gadgets	GizmoWorks
Powergizmo	$29.99	Gadgets	GizmoWorks
SingleTouch	$149.99	Photography	Canon
MultiTouch	$203.99	Household	Hitachi

```
SELECT   DISTINCT manufacturer
FROM     Product
WHERE    category ='Photography'
OR       category ='Household'
```



Manufacturer
Canon
Hitachi

Combining Predicates

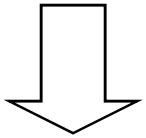
Show all manufacturer that make both photography and household items

Product

PName	Price	Category	Manufacturer
Gizmo	$19.99	Gadgets	GizmoWorks
Powergizmo	$29.99	Gadgets	GizmoWorks
SingleTouch	$149.99	Photography	Canon
MultiTouch	$203.99	Household	Hitachi

WRONG!

```
SELECT   DISTINCT maker
FROM     Product
WHERE    category ='Photography'
AND      category ='Household'
```



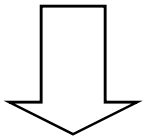
PName	Price	Category	Manufacturer

Combining Predicates

Product

PName	Price	Category	Manufacturer
Gizmo	$19.99	Gadgets	GizmoWorks
Powergizmo	$29.99	Gadgets	GizmoWorks
SingleTouch	$149.99	Photography	Canon
MultiTouch	$203.99	Household	Hitachi

```
SELECT  *
FROM    Product
WHERE   pname LIKE '%Giz%'
OR      category LIKE '%hold'
AND     price > 100
```



AND has higher precedence over OR

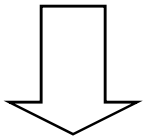
Pname	Price	Category	Manufacturer
Gizmo	$19.99	Gadgets	GizmoWorks
MultiTouch	$203.99	Household	Hitachi

Combining Predicates

Product

PName	Price	Category	Manufacturer
Gizmo	$19.99	Gadgets	GizmoWorks
Powergizmo	$29.99	Gadgets	GizmoWorks
SingleTouch	$149.99	Photography	Canon
MultiTouch	$203.99	Household	Hitachi

```
SELECT  *
FROM    Product
WHERE   (pname LIKE '%Giz%'
OR       category LIKE '%hold')
AND     price > 100
```



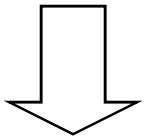
Pname	Price	Category	Manufacturer
MultiTouch	$203.99	Household	Hitachi

Ordering the Results

Default order is Ascending
Use DESC for descending

PName	Price	Category	Manufacturer
Gizmo	$19.99	Gadgets	GizmoWorks
Powergizmo	$29.99	Gadgets	GizmoWorks
SingleTouch	$149.99	Photography	Canon
MultiTouch	$203.99	Household	Hitachi

```
SELECT   pName, price, category, manufacturer
FROM     Product
WHERE    price > 50
ORDER BY pName
```



Pname	Price	Category	Manufacturer
MultiTouch	$203.99	Household	Hitachi
SingleTouch	$149.99	Photography	Canon

ORDER BY: Sorting the Results

```
SELECT   PName, Price, Manufacturer
FROM     Product
WHERE    category='Gadgets' AND price > 10
ORDER BY price, pName
```

Ties are broken by the second attribute on the ORDER BY list, etc.

Ordering is ascending, unless you specify the DESC keyword.

ORDER BY: independent ordering possible

```
SELECT    PName, Price, Manufacturer
FROM      Product
WHERE     category='Gadgets' AND price > 10
ORDER BY  price ASC, pName DESC
```

NULL values always comes last
You can say
…
ORDER BY price ASC NULLS FIRST
…
ORDER BY price ASC NULLS LAST

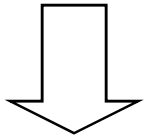
IN operator

Find all albums launched in 1984 or 1996.

CDCollection

artist	album	tracks	company	year
U2	The Unforgettable	10	Island	1984
U2	Rattle and Hum	17	Island	1988
U2	Achtung Boys	12	Island	1991
Underworld	Second Toughest	8	Junior	1996

```
SELECT  album
FROM    CDCollection
WHERE   year IN (1984, 1996)
```



```
SELECT  album
FROM    CDCollection
WHERE   year =1984
OR      year =1996
```

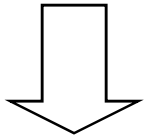
artist	album	tracks	company	year
U2	The Unforgettable	10	Island	1984
Underworld	Second Toughest	8	Junior	1996

NOT IN operator

CDCollection

artist	album	tracks	company	year
U2	The Unforgettable	10	Island	1984
U2	Rattle and Hum	17	Island	1988
U2	Achtung Boys	12	Island	1991
Underworld	Second Toughest	8	Junior	1996

```
SELECT  *
FROM    CDCollection
WHERE   year NOT IN (1984, 1996)
```



```
SELECT  *
FROM    CDCollection
WHERE   year < > 1984
AND     year < > 1996
```

artist	album	tracks	company	year
U2	Rattle and Hum	17	Island	1988
U2	Achtung Boys	12	Island	1991